LELEC2870 Machine Learning – Project Intermediate Report: Part 1

Linear Baseline and Feature Selection for Smurf Heart Failure Risk Prediction

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1. Problem Formulation & Preprocessing

Part 1 aims to predict 10-year heart failure risk $y \in [0, 1]$ using clinical data, constrained to selecting **at most 6 original features**.

Target Exploration: The training distribution ($n = 1000$) is positively skewed (mean 0.081, median 0.060, max 0.490). While 69.5% of Smurfs are low-risk ($y < 0.10$), 30.5% exceed Doctor Smurf's critical clinical threshold ($y \geq 0.10$), where sample weights $w_i \in [2.0, 5.0]$ heavily penalize under-prediction.

Preprocessing Pipeline: To eliminate data leakage, all transformations are encapsulated in a `ColumnTransformer` fitted strictly on training folds:

1. *Metadata removal:* The non-predictive MRI scan pointer `img_filename` is dropped.
2. *Categorical encoding:* Consumption features (`sarsaparilla`, `smurfberry_liquor`, `smurfin_donuts`) display monotonic severity and are ordinally mapped (0 to 4) and standardized. `profession` is nominal and encoded via One-Hot encoding (`drop='first'`).
3. *Feature scaling:* Numerical predictors are standardized ($\mu = 0, \sigma = 1$).

2. Feature Selection (≤ 6 Features)

To select the optimal 6-feature subset without bias, we benchmarked the three core families from TP3: *Filters*, *Wrappers*, and *Embedded* methods.

Method	Selected Features Subset	CV RW-RMSE
Corr. Filter	BP, Chol, Weight, Donuts, Sarsa, Age	0.0623
Mutual Info	BP, Weight, Donuts, Chol, Potassium, Sarsa	0.0631
mRMR	BP, Potassium, Craftsmanship, Donuts, Hb, Height	0.0645
SFS Forward	**BP, Weight, Hb, Height, Sarsa, Age**	**0.0598**
SFS Backward	**BP, Weight, Hb, Height, Sarsa, Age**	**0.0598**
LassoCV (α^*)	BP, Weight, Age, Sarsa, Hb, Chol	0.0602

Table 1: Comparison of feature selection paradigms (5-fold CV).

Analysis & Selected Subset:

- *Unanimous core:* `blood_pressure` (BP, $r = +0.69$), `weight` ($r = +0.41$), `age` ($r = +0.21$), and `sarsaparilla` (Sarsa, $r = -0.24$) appear in every predictive model.
- *The Hemoglobin (Hb) discovery:* While simple correlation favors `smurfin_donuts` ($r = +0.37$), SFS and Lasso select `hemoglobin` ($r = -0.16$). Pairwise analysis proves that donut consumption shares redundant collinear variance with `weight`, whereas `hemoglobin` is virtually orthogonal to all other predictors ($|\rho| \leq 0.041$), introducing unique, uncaptured physiological variance.
- *Tie-breaking the 6th slot:* Wrappers favor `height` (CV RW-RMSE 0.05976) to proxy Body Mass Index (W/H), whereas Lasso favors `cholesterol` (CV RW-RMSE 0.06019). With that, we chose `cholesterol` on grounds of medical interpretability.

Final 6 Features: {`blood_pressure`, `weight`, `age`, `sarsaparilla`, `hemoglobin`, `cholesterol`}.

3. Model Formulation: WLS vs. OLS

Standard OLS minimizes unweighted squared error $\sum(y_i - \hat{y}_i)^2$, treating a 1% error on a healthy Smurf identically to a dying one.

Because RW-RMSE weights errors by $w_i = 1 + y_i/0.1$ and depends **strictly on the ground truth**, we directly align our objective with the metric via **Weighted Least Squares (WLS)**:

$$\hat{\boldsymbol{\beta}}_{\text{WLS}} = (\mathbf{X}^T\mathbf{W}\mathbf{X})^{-1}\mathbf{X}^T\mathbf{W}\mathbf{y}, \quad \mathbf{W} = \text{diag}(w_1, \ldots, w_n) \tag{1}$$

4. Experimental Results & Generalization

We evaluated models using 5-fold cross-validation on X_{train} ($n = 1000$), holding out X_{test} ($n = 500$) untouched until final evaluation.

Model / Architecture	Train RW	Val RW-RMSE	Val RMSE
Naive Baseline ($\bar{y}$)	0.1099	0.1099 ± 0.0042	0.0815
OLS (Unweighted)	0.0622	0.0628 ± 0.0054	**0.0514**
Ridge WLS ($\alpha = 1.0$)	0.0592	0.0602 ± 0.0039	0.0536
Lasso WLS ($\alpha = 0.0006$)	0.0593	0.0602 ± 0.0040	0.0536
WLS Baseline (Selected)	**0.0592**	$\mathbf{0.0602 \pm 0.0039}$	0.0536

Table 2: Cross-validation performance on X_{train} ($n = 1000$).

Key Empirical Findings:

1. *WLS Supremacy:* WLS delivers a $+4.18$% **gain** in validation RW-RMSE over OLS. OLS achieves lower unweighted RMSE (0.0514 vs 0.0536) by sacrificing accuracy on the high-risk tail, proving that optimizing unweighted MSE is suboptimal for Doctor Smurf.
2. *Regularization:* With $n = 1000$ and only $p = 6$ standardized features, L_1/L_2 shrinkage (Ridge/Lasso) produces negligible variance reduction; pure WLS remains optimal and parsimonious.
3. *Generalization on X_{test}:* Evaluated once on the unseen test set ($n = 500$), our model achieves an **RW-RMSE of** 0.06621 and **RMSE of** 0.05666, representing a 39.7% **error reduction** over the naive mean baseline (0.10987).
4. *Learned Coefficients:* Standardized weights demonstrate that `blood_pressure` is the leading risk driver ($\beta = +0.0481$), followed by protective effects from `hemoglobin` ($\beta = -0.0159$) and `sarsaparilla` ($\beta = -0.0130$), while `age` ($+0.0120$), `cholesterol` ($+0.0119$), and `weight` ($+0.0117$) contribute equally.

5. Deliverable Verification & Part 2 Outlook

Submission Quality: For the intermediate delivery, the model was refit on all labeled data ($n = 1500$) with sample weights and applied to $X_{\text{unlabeled}}$ ($n = 500$). `y_pred.csv` was verified: exactly 500 rows, no headers/index, values in $[0.000, 0.459]$ (mean 0.0998), with 18 unphysical negative values cleanly clipped to zero.

Limitations & Transition to Part 2: While linear WLS offers a robust benchmark, its limitations are evident:

- *Non-linear saturations:* BP and cholesterol likely exhibit exponential risk escalations above clinical thresholds.
- *Feature interactions:* Synergistic effects (e.g. high cholesterol and BP) require explicit polynomial terms.